Dynamically Scheduled Instruction Processing: Tomasulo

Problems?

- **How do we prevent WAR and WAW hazards?**
- **How do we deal with variable latency?**

Instruction		Clock Cycle Number																
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
LD	F6,34(R2)	IF	ID	EX	MEM	WB												
LD	F2,45(R3)		IF	ID	EX	MEM	WB											
MULTD	F0,F2,F4			IF	ID	stall	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	MEM	WB
SUBD	F8,F6,F2				IF	ID	A1	A2	MEM	WB								
DIVD	F10,F0,F6					IF	ID	stall	stall	stall	stall	stall	stall	stall	stall	stall	D1	D2
ADDD	F6,F8,F2						IF	ID	A1	A2	MEM	WB						

RAW

WAR

What do registers offer?

- **Short, absolute name for a recently computed (or frequently used) value**
- **Fast, high bandwidth storage in the data path**
- **Means of broadcasting a computed value to set of instructions that use the value**

Broadcasting result value

- **Series of instructions issued and waiting for value to be produced by logically preceding instruction.**

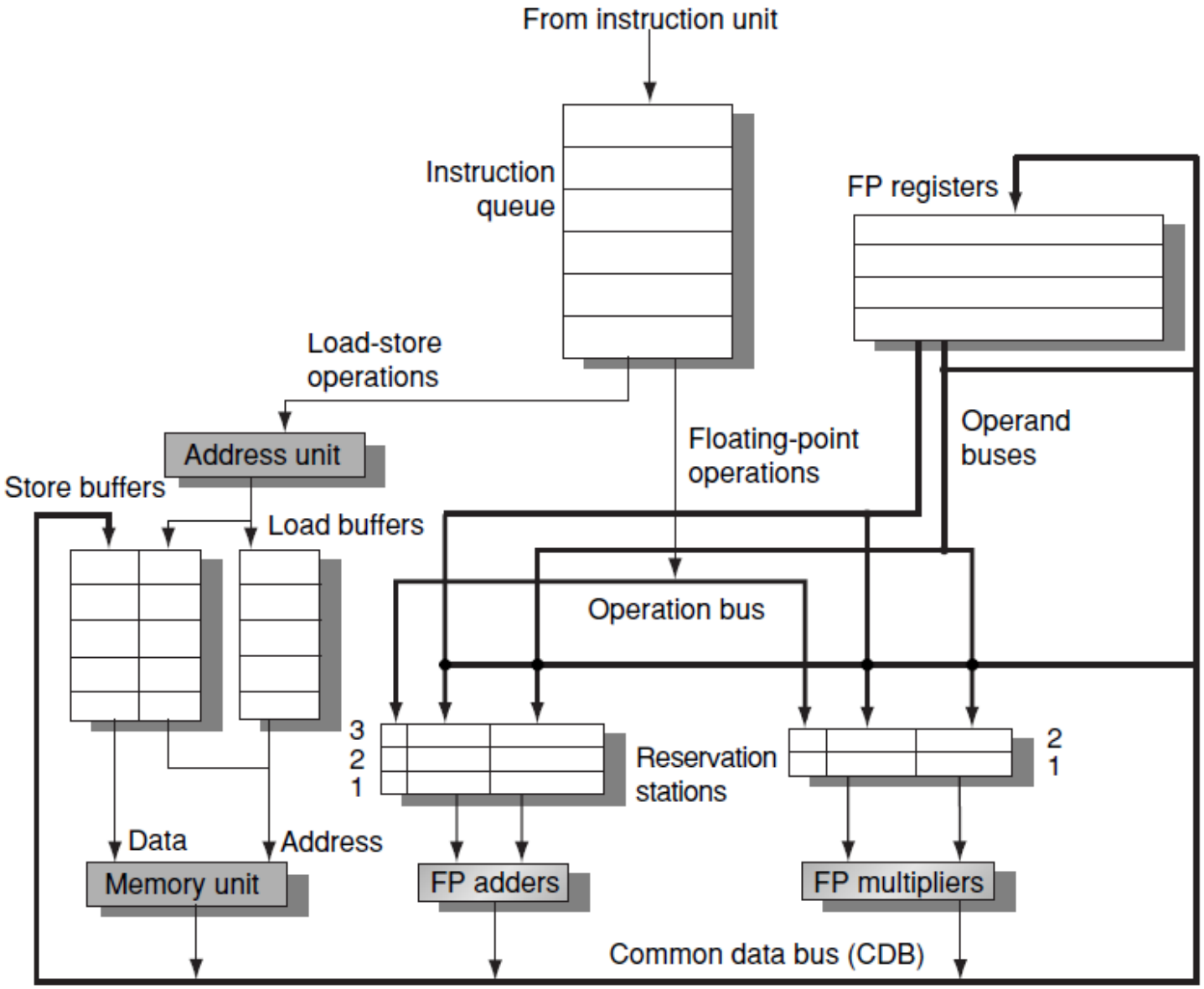
- **CDC6600 has each come back and read the value once it is placed in register file**

- **Alternative: broadcast value and reg # to all the waiting instructions**

Tomasulo Algorithm vs. Scoreboard

- Control & buffers distributed with Function Units (FU) vs. centralized in scoreboard;
  - FU buffers called "reservation stations"; have pending operands
- Registers in instructions replaced by values or pointers to reservation stations (RS); called register renaming ;
  - avoids WAR, WAW hazards
  - More reservation stations than registers, so it can do optimizations compilers can't
- Results to FU from RS, not through registers. Over Common Data Bus that broadcasts results to all FUs
- Load and Stores treated as FUs with RSs as well

The basic structure of a MIPS floating-point unit using Tomasulo's algorithm



Load and store buffer

- **Load buffer**
  - Hold the components of the effective address until it is computed
  - Track outstanding loads that are waiting on the memory
  - Hold the results of completed loads

- **Store buffer**
  - Hold the components of the effective address until it is computed
  - Hold the destination memory addresses of outstanding stores that are waiting for data value to store
  - Hold the address and value to store until the memory unit is available

Three Stages of Tomasulo Algorithm

Each stage could take an arbitrary number of clock cycles.

1. Issue stage—get instruction from FP Op Queue

- **FIFO order to ensure the correct data flow**
- **If reservation station free (no structural hazard), control issues instr & sends operands (renames registers).**
- **If no empty reservation: structural hazard, stall until a station or buffer is freed**
- **if operands are not in the registers, keep track of the functional units that will produce the operands.**
- **Eliminate WAR and WAW hazards**

Execution Stage of Tomasulo Algorithm

2. Execution—operate on operands (EX)

- If both operands ready then execute;
- if not ready, watch Common Data Bus for result (Resolve RAW)
- several instructions could become ready in the same clock cycle
- independent functional units could begin execution in the same clock cycle for different instructions
- if more than one instruction is waiting for a single functional unit, the unit will choose among them.

Write result Stage of Tomasulo Algorithm

3. **Write result —finish execution (WB)**

   Write on Common Data Bus to all awaiting units; mark reservation station available

- **Normal data bus: data + destination ("go to" bus)**
- **Common data bus: data + source ("come from" bus)**
  - **64 bits of data + 4 bits of Functional Unit source address**
  - **Write if matches expected Functional Unit (produces result)**
  - **Does the broadcast**

Reservation Station : Seven Fields

Op: Operation to perform in the unit

Vj, Vk: Value of Source operands

- Store buffers has V field, result to be stored

Qj, Qk: Reservation stations producing source registers (source registers to be written by other instructions)

- Note: Qj, Qk=0 indicates source is already in Vj or Vk (or is unnecessary)

Busy: Indicates reservation station or FU is busy

A: Hold information for memory address calculation for a load store

- initially: immediate field of the instruction
- after address calculation: effective address

Register result status: one field Qi

- **Indicates the number of the reservation station (which functional unit) will write the register, if one exists.**

- **Blank when no pending instructions that will write that register.**

Load and store buffers each have a field : A

- **Holds the result of the effective address once the first step of execution has been completed**

Notations

- r : Reservation Station or buffer that the instruction is assigned to
- rs, rt : source register number
- rd : destination register number
- RS : Reservation Station
- RegStat : Register status data structure
- Regs : Register file

Tomasulo's Algorithm: The Details

Instruction state	Wait until	Action/bookkeeping

Issue
ALU operation
	Reservation Station r empty	if (RegStat[rs].Qi ≠ 0)
		{RS[r].Qj ← RegStat[rs].Qi}
		else { RS[r].Vj ← Regs[rs]; RS[r].Qj ←0}
		if (RegStat[rt].Qi ≠ 0)
		{RS[r].Qk ← RegStat[rt].Qi}
		else { RS[r].Vk ← Regs[rt]; RS[r].Qk ←0}
		RS[r].busy ← yes;
		RegStat[rd].Qi ← r;

Issue
Load or store
	Buffer r empty	if (RegStat[rs].Qi ≠ 0)
		{RS[r].Qj ← RegStat[rs].Qi}
		else { RS[r].Vj ← Regs[rs]; RS[r].Qj ←0}
		RS[r].A ← imm;
		RS[r].busy ← yes;
		RegStat[rt].Qi ← r; // for load only
		If (RegStat[rt].Qi ≠ 0) // for store only
		{RS[r].Qk ← RegStat[rt].Qi}
		Else {RS[r].Vk ←Regs[rt]; RS[r].Qk ←0 }

Tomasulo's Algorithm: The Details

Instruction state	Wait until	Action/bookkeeping
Execute ALU operation	RS[r].Qj=0 and RS[r].Qk=0	Compute result: operands are in Vj and Vk
Execute Load-store step 1	RS[r].Qj=0 & r is head of load-store queue	RS[r].A ← RS[r].Vj + RS[r].A;
Execute Load step 2	Load step1 complete	Read from Mem[RS[r].A]

Tomasulo's Algorithm: The Details

Instruction state	Wait until	Action/bookkeeping
Write Result ALU operation or load	Execution complete at r & CDB available	∀ x (if(RegStat[x].Qi=r)) { Regs[x]←result ; RegStat[x].Qi←0} ∀ x (if(RS[x].Qj=r)) { RS[x].Vj←result ; RS[x].Qj←0} ∀ x (if(RS[x].Qk=r)) { RS[x].Vk←result ; RS[x].Qk←0} RS[r].busy ← no;
Write Result Store	Execution complete at r & RS[r].Qk=0	Mem[RS[r].A] ← RS[r].Vk; RS[r].busy ← no;

Tomasulo Example

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2			
LD	F2	45+	R3			
MULTD	F0	F2	F4			
SUBD	F8	F6	F2			
DIVD	F10	F0	F6			
ADDD	F6	F8	F2			

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
	Mult2	No					

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
0	*FU*									

Tomasulo Example Cycle 1

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1		
LD	F2	45+	R3			
MULTD	F0	F2	F4			
SUBD	F8	F6	F2			
DIVD	F10	F0	F6			
ADDD	F6	F8	F2			

	Busy	*Address*
Load1	Yes	34+R2
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
	Mult2	No					

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
1	*FU*				Load1					

Tomasulo Example Cycle 2

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1		
LD	F2	45+	R3	2		
MULTD	F0	F2	F4			
SUBD	F8	F6	F2			
DIVD	F10	F0	F6			
ADDD	F6	F8	F2			

	Busy	Address
Load1	Yes	34+R2
Load2	Yes	45+R3
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
	Mult2	No					

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
2	*FU*		Load2		Load1					

Tomasulo Example Cycle 3

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	
LD	F2	45+	R3	2		
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2			
DIVD	F10	F0	F6			
ADDD	F6	F8	F2			

	Busy	Address
Load1	Yes	34+R2
Load2	Yes	45+R3
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	Yes	MULTD		R(F4)	Load2	
	Mult2	No					

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
3	*FU*	Mult1	Load2		Load1					

- **Note: registers names are removed ("renamed") in Reservation Stations;**
- **Load1 completing; what is waiting for Load1?**

Tomasulo Example Cycle 4

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4		
DIVD	F10	F0	F6			
ADDD	F6	F8	F2			

	Busy	Address
Load1	No	
Load2	Yes	45+R3
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	Yes	SUBD	M(A1)			Load2
	Add2	No					
	Add3	No					
	Mult1	Yes	MULTD		R(F4)	Load2	
	Mult2	No					

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
4	*FU*	Mult1	Load2		M(A1)	Add1				

- **Load2 completing; what is waiting for Load2?**

Tomasulo Example Cycle 5

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4		
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2			

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
2	Add1	Yes	SUBD	M(A1)	M(A2)		
	Add2	No					
	Add3	No					
10	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
5	*FU*	Mult1	M(A2)		M(A1)	Add1	Mult2			

Tomasulo Example Cycle 6

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4		
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6		

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
1	Add1	Yes	SUBD	M(A1)	M(A2)		
	Add2	Yes	ADDD		M(A2)	Add1	
	Add3	No					
9	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
6	*FU*	Mult1	M(A2)		Add2	Add1	Mult2			

Tomasulo Example Cycle 7

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4	7	
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6		

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
0	Add1	Yes	SUBD	M(A1)	M(A2)		
	Add2	Yes	ADDD		M(A2)	Add1	
	Add3	No					
8	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
7	*FU*	Mult1	M(A2)		Add2	Add1	Mult2			

- **Add1 completing; what is waiting for it?**

Tomasulo Example Cycle 8

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6		

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
2	Add2	Yes	ADDD	(M-M)	M(A2)		
	Add3	No					
7	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
8	*FU*	Mult1	M(A2)		Add2	(M-M)	Mult2			

Tomasulo Example Cycle 9

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6		

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
1	Add2	Yes	ADDD	(M-M)	M(A2)		
	Add3	No					
6	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
9	*FU*	Mult1	M(A2)		Add2	(M-M)	Mult2			

Tomasulo Example Cycle 10

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6	10	

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
0	Add2	Yes	ADDD	(M-M)	M(A2)		
	Add3	No					
5	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
10	*FU*	Mult1	M(A2)		Add2	(M-M)	Mult2			

- **Add2 completing;**

Tomasulo Example Cycle 11

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
4	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
11	*FU*	Mult1	M(A2)		(M-M+M	(M-M)	Mult2			

- **Write result of ADDD here vs. scoreboard?**
- **All quick instructions complete in this cycle!**

Tomasulo Example Cycle 12

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
3	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
12	*FU*	Mult1	M(A2)		(M-M+M	(M-M)	Mult2			

Tomasulo Example Cycle 13

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
2	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
13	*FU*	Mult1	M(A2)		(M-M+M	(M-M)	Mult2			

Tomasulo Example Cycle 14

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3		
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
1	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
14	*FU*	Mult1	M(A2)		(M-M+M	(M-M)	Mult2			

Tomasulo Example Cycle 15

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3	15	
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
0	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
15	*FU*	Mult1	M(A2)		(M-M+M	(M-M)	Mult2			

Tomasulo Example Cycle 16

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3	15	16
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
40	Mult2	Yes	DIVD	M*F4	M(A1)		

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
16	*FU*	M*F4	M(A2)		(M-M+M	(M-M)	Mult2			

Faster than light computation (skip a couple of cycles)

Tomasulo Example Cycle 55

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3	15	16
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5		
ADDD	F6	F8	F2	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
1	Mult2	Yes	DIVD	M*F4	M(A1)		

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
55	*FU*	M*F4	M(A2)		(M-M+M	(M-M)	Mult2			

Tomasulo Example Cycle 56

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3	15	16
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5	56	
ADDD	F6	F8	F2	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
0	Mult2	Yes	DIVD	M*F4	M(A1)		

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
56	*FU*	M*F4	M(A2)		(M-M+M	(M-M)	Mult2			

- **Mult2 is completing; what is waiting for it?**

Tomasulo Example Cycle 57

Instruction status:

Instruction		*j*	*k*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	3	4
LD	F2	45+	R3	2	4	5
MULTD	F0	F2	F4	3	15	16
SUBD	F8	F6	F2	4	7	8
DIVD	F10	F0	F6	5	56	57
ADDD	F6	F8	F2	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Reservation Stations:

Time	*Name*	*Busy*	*Op*	*S1 Vj*	*S2 Vk*	*RS Qj*	*RS Qk*
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
	Mult2	Yes	DIVD	M*F4	M(A1)		

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
56	*FU*	M*F4	M(A2)		(M-M+M	(M-M)	Result			

- **Once again: In-order issue, out-of-order execution and completion.**

Compare to Scoreboard Cycle 62

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*	*Issue*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4	1	3	4
LD	F2	45+	R3	5	6	7	8	2	4	5
MULTD	F0	F2	F4	6	9	19	20	3	15	16
SUBD	F8	F6	F2	7	9	11	12	4	7	8
DIVD	F10	F0	F6	8	21	61	62	5	56	57
ADDD	F6	F8	F2	13	14	16	22	6	10	11

Why is Tomasulo's scheme widely adopted in multiple-issue processors after 1990s

- Achieving high performance without requiring the compiler to target code to a specific pipeline structure.
- With the presence of cache, the delays are unpredictable (miss). Out-of-order execution allows the processors to continue executing instructions while awaiting the completion of a cache miss (hiding cache miss penalty).
- Processors becoming more aggressive in issue capability demands register renaming and dynamic scheduling.
- Dynamic scheduling is a key component of speculation, it was adopted along with hardware speculation in the mid-1990s.

Hardware-Based Speculation

- **Speculation: fetch, issue, and execute instructions as if the branch predictions were always correct**
- **key ideas of Hardware-based speculation**
  - **Dynamic branch prediction**
  - **Allow execution of instructions before the control dependences are resolved**
  - **Dynamic scheduling of different combinations of basic blocks**
- **Perform update that cannot be undone only when the instruction is no longer speculative**
  - **Update the register file or memory: instruction commit**
- **Execute out of order but commit in order**

Hardware-Based Speculation

- **Allow instruction to execute out of order but commit in order**
- **Separate execution complete from instruction commit**
- **Require additional hardware buffers (Reorder buffer, ROB) : hold the results (finished execution but have not committed)**

- **Tomasulo's algorithm with speculation**
  - **4 steps**
  - **Issue, execute, write result, commit**

Differences between Tomasulo's algorithm with and without speculation

- **Without speculation**
  - **Only partially overlaps basic blocks because it requires that a branch to be resolved before actually executing any instructions in the successor basic block**
  - **Once an instruction writes its result, any subsequently issued instructions will find the result in the register file**

- **With speculation**
  - **Dynamic scheduling of different combinations of basic blocks**
  - **The register file is not updated until the instruction commits**
  - **integrate the function of store buffer into ROB**

Four steps of hardware-based speculation

1. Issue

- Get the instruction from the instruction queue
- Issue if there is an empty reservation and empty slot in ROB (stall if no empty entry)
- Send the operands to reservation station if they are available
- Update the control entries
- The number of the ROB entry allocated for the result is also sent to the reservation station

2. Execute

- If one or more operands is not yet available, monitor the CDB
- When both operands are available at a reservation station, execute the operation (eliminate RAW)

Four steps of hardware-based speculation

3. Write result

- When the result is available, write it on the CDB , any reservation stations waiting for this result, and from the CDB to ROB

4. Commit

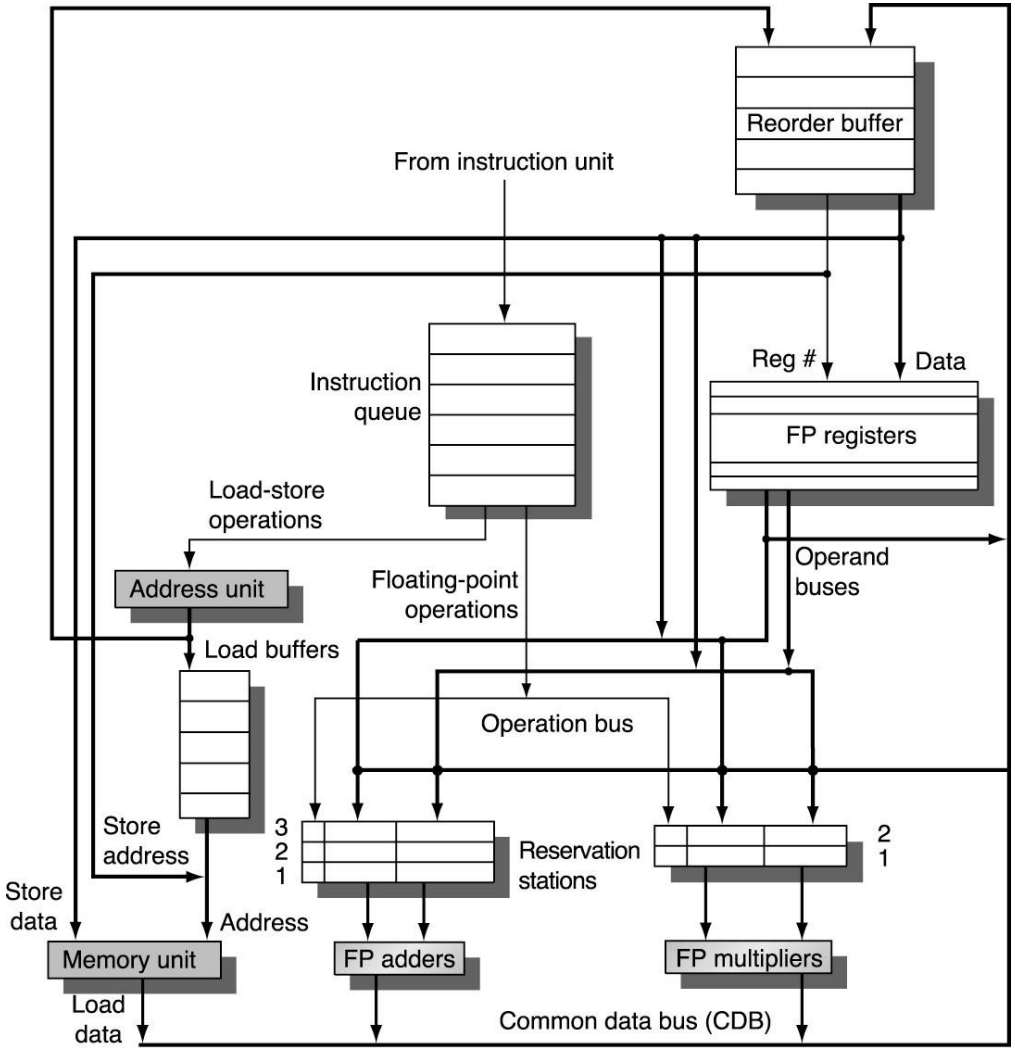
- Branch with an incorrect prediction: the speculation is wrong, so the ROB is flushed and the execution is restarted at the correct successor of the branch

  (if the branch is correctly predicted, the branch is finished)

- Store: when an instruction reaches the head of ROB, update the memory with the result and removes the instruction from the ROB

- Any other instruction: when an instruction reaches the head of ROB, update the register with the result and removes the instruction from the ROB

Reorder Buffer



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